STRUCTURAL MODELLING & SIMULATION ENGINEER — PARIS

Rahul Subramanian *Girija*

I model how aerospace structures carry load — finite-element simulation in Abaqus, geometry in CATIA — on the foundation of a PhD in materials science. And I'm automating my FEA pipeline with Python and Claude Code.

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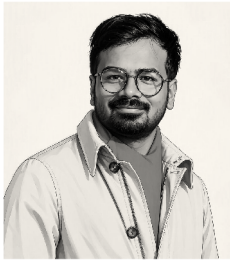


Fig. 1

FOCUS	TOOLS	AUTOMATION	DOCTORATE
Structural FE simulation	Abaqus · CATIA · Python	AI-assisted FEA pipelines	Materials Science, I2M / CNRS

Work experience

09/2026 — Present
Paris, France

Structural Modelling and Simulation Engineer
Design Tech Centre — Paris

- **Structural analysis:** static, nonlinear, buckling, modal, and thermal finite-element analyses in Abaqus, working from CATIA geometry.
- **Design to specification:** responding to customer specifications (*cahiers des charges*) for the mechanical design of aircraft components — mostly small, innovative light aircraft powered by new energy sources.
- **Automated FEA pipeline (Python + Claude Code):** problem definition, CAD generation, Abaqus input files (.inp) written through Abaqus/CAE, parametric studies, and data post-processing — through to automatically built slide decks and interactive viewers of the simulated product.
- **Prompt to product:** an end-to-end digital design workflow that goes from a written prompt to a thoroughly analysed product, delivered in an interactive viewer.

04/2024 — 09/2026
Toulouse, France

Mechanical and Physical Engineer
Capgemini Engineering — embedded at Liebherr Aerospace Toulouse

Scientific Specialist, Materials & Special Processes

- **Compact aluminium heat exchangers (Next Single Aisle):** solved a brazing-joint quality issue on 80 µm fins by tracing clad diffusion through extensive microstructural analysis — findings that led to a change in Liebherr's fin-forming process.
- **Graphite seal rings for valves:** built an ASTM-compliant material characterization framework and supplier qualification protocol, giving the team an end-to-end path to evaluate and select vendors.
- **Inconel 718 valve body redesign:** led microstructural characterization across casting, forging, and additive manufacturing routes, linking microstructure to post-heat-treatment properties to guide material selection for high-temperature service.

02/2020 — 08/2023
Bordeaux, France

PhD / Young Researcher
I2M, National Centre for Scientific Research (CNRS)

With Institut P', LASIE & MCAM (Australia)

- Investigated hydrogen embrittlement in laser powder-bed-fusion 316L, 304L, and 17-4PH stainless steels under tensile and high-cycle fatigue loading.
- Designed and built a coupled fatigue–hydrogen electrochemical testing rig, running high-pressure hydrogen gas mechanical tests as part of a three-lab collaboration.
- Fabricated, machined, and characterized specimens, communicated results regularly, and mentored Master's students and interns on the project.

Skills & technical knowledge

NUMERICAL & SIMULATION	Abaqus / OIM Analysis / ATEX / FEM / Zebulon / Meshlab / Gmsh / AutoCAD / CATIA / Autodesk Inventor / Python / Claude Code / MATLAB / Materialise MAGICS / Avizo
DOMAIN EXPERTISE	Structural Simulation (FEA) / FEA Automation / Nonlinear Analysis / Buckling & Modal Analysis / Thermal Analysis / Parametric Studies / Mechanical Design / Materials Science / Metal Additive Manufacturing / Fatigue of Metals / Corrosion / Tribology / Hydrogen Embrittlement / Heat Exchangers / Process Optimization / Experimental Investigation / Project Management / Supplier Evaluation / Market Research / Scientific Research

CHARACTERIZATION	Microscopy / SEM / TEM / XRD / EBSD / EDS / X-ray Tomography / AFM / TDS / GD-OES / Mechanical Testing / Alicona / Keyence
SOFT SKILLS & TOOLS	Scientific Writing / Technical Presentation / Problem Solving / Team Work / Listening / Transparency / Empathy / Origin Pro / Office 365 / LaTeX
LANGUAGES	**English** — Native / Bilingual / **French** — Full Professional

Education

02/2020 – 02/2024
Bordeaux, France

PhD — Materials Science
HESAM University · ENSAM · Institut de Mécanique et d'Ingénierie (I2M)

Doctoral research on hydrogen embrittlement in additively manufactured stainless steels, conducted in collaboration with Institut P', LASIE, and MCAM (Australia).

09/2017 – 09/2019
Poitiers, France

Master — Aeronautical Mechanics & Energetics (AME)
ISAE-ENSMA — École Nationale Supérieure de Mécanique et d'Aérotechnique

Specialisation in high-temperature materials. HIGH-TEMPERATURE MATERIALS TRACK

Scientific projects

04/2019 – 10/2019

Master-2 Research Internship · Institut P' / UGITECH
Assessed hydrogen embrittlement in stainless steels at low temperatures.

09/2018 – 02/2019

Master's Research Project · Safran Aircraft Engines
Studied the effect of over-aging on the creep behaviour of cast Inconel 718 alloy.

04/2018 – 10/2018

Master-1 Research Internship · Institut P' / ANR ImPEKKable
Characterized nanoindentation creep behaviour of epoxy, aged epoxy, and PEKK polymer at high temperatures.

11/2017 – 07/2018
ARIANE GRAND PRIZE

Cryogenic Liquid Oxygen Tank — Team Project · Student Aerospace Challenge '18
Studied heat transfer modes and material choices to reduce vaporisation losses and improve tank design. Awarded the Ariane Grand Prize for Best Project.

11/2016 – 04/2017

Acoustic Emission in Composite Laminates · Bachelor Final Year Internship
Investigated the acoustic emission behaviour of glass-fibre composite laminates subject to small-scale impact.

Publications

Conference Proceedings · 2022

Assessment of the influence of hydrogen on the fatigue life of selective laser melted 316L austenitic stainless steels
Subramanian Girija, et al. — with Nicolas Saintier, Gilbert Hénaff, Isabelle Aubert, Mohamed El May, Abdelali Oudriss, Xavier Feaugas

International Conference on Metals and Hydrogen — Steelyhydrogen

Certificates

06/2025 – 07/2025

Emerging Leaders Program — Grade B
Leadership development program offered by Capgemini in collaboration with Harvard Business Review.

08/2025 – 09/2025

Sustainability for Products & Systems Engineers
Beginner course on sustainability, systems thinking, eco-design, life-cycle analysis, and sustainable energy sources.